star’s implosion simultaneously from the viewpoint of faraway static observers and from the viewpoint of observers who ride inward with the imploding star. The resulting description reconciled beautifully the freezing of the implosion as observed from far away with the continued implosion as observed from the star’s surface.

In 1962, two members of Wheeler’s Princeton research group, David Beckedorff and Charles Misner, constructed a set of embedding diagrams to illustrate this reconciliation, and in 1967 I converted their embedding diagrams into the following fanciful analogy for an article in *Scientific American.*

Once upon a time, six ants lived on a large rubber membrane (Figure 6.6). These ants, being highly intelligent, had learned to communicate using signal balls that roll with a constant speed (the “speed of light”) along the membrane’s surface. Regrettably, the ants had not calculated the membrane’s strength.

One day five of the ants happened to gather near the center of the membrane, and their weight made it begin to collapse. They were trapped; they could not crawl out fast enough to escape. The sixth ant—an astronomer ant—was a safe distance away with her signal-ball telescope. As the membrane collapsed, the trapped ants dispatched signal balls to the astronomer ant so she could follow their fate.

The membrane did two things as it collapsed: First, its surface contracted inward, dragging surrounding objects toward the center of the collapse in much the same manner as an imploding star’s gravity pulls objects toward its center. Second, the membrane sagged and became curved into a bowl-like shape analogous to the curved shape of space around an imploding star (compare with Figure 6.2).